

Coding & Solution

Date	15 July 2026
Team ID	XXXXXX
Project Name	AI Debt Settlement System
Maximum Marks	5 Marks

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Solution Summary

Field	Details
Repository Link / URL	AIndhu01/AI-Powered-Debt-Relief-Financial-Recovery-Platform/tree/main -Project-Template
Programming Language(s)	Python, JavaScript, HTML5, CSS3
Framework(s) Used	FastAPI, React.js, Tailwind CSS, SQLAlchemy, Uvicorn
Key Features Implemented	• User Registration & Login (JWT Authentication) • Loan Management (Add, Edit, Delete, Search, History) • Dashboard Analytics (Financial Health Score, Debt Stress, EMI Ratio, Monthly Surplus) • AI-Based Debt Settlement Recommendation using Google Gemini AI • User Profile with Total Loans & Total Debt • REST API Integration between React and FastAPI
Pending / Incomplete Features	• Cloud Deployment (Render/Vercel) • Email Notifications • Admin Dashboard • AI Chatbot for Financial Assistance (<i>Future Enhancement</i>)
Setup / Run Instructions	Backend: cd backend → python pip install -r requirements.txt → python -m uvicorn main:app --reload Frontend: cd frontend → npm install → npm run dev Access the application at http://localhost:5173

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions/classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments/documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

- The project follows a **three-tier architecture** consisting of a React frontend, FastAPI backend, and SQLite database.
- JWT Authentication is implemented to ensure secure access to protected resources.
- Google Gemini AI is integrated to generate intelligent debt settlement recommendations based on the user's financial information.
- SQLAlchemy ORM is used for efficient database management and CRUD operations.
- The application provides a responsive user interface with Tailwind CSS for an enhanced user experience.
- The project is designed to be scalable and can be extended with features such as cloud deployment, admin management, predictive analytics, and AI-powered financial chat assistance in future versions.
- The complete source code is maintained in a GitHub repository using Git for version control and collaborative development.